

10.5
10

Pre Lab 7

Alex Le

February 2026

1 Doc Statement

None

2 'Doc' Statement



H*H*H
+0.5

3 Reaction Wheel Requirements

3.1 Total Torque To Counter Disturbances

Gravity Gradient:

$$T_{\text{grav}} = \frac{3\mu}{2R^3} |I_{\text{yaw}} - I_{\text{other}}| \sin(2\theta)$$

$$\dots = \frac{3 \cdot 3.986005 \cdot 10^{14} \frac{\text{m}^3}{\text{s}^2}}{2 \cdot (6578000)^3 \text{m}^3} \cdot |0.014 - 0.010| \text{kg} \cdot \text{m}^2 \sin(2 \cdot 45^\circ)$$

$$\dots = 8.4025 \cdot 10^{-9} \text{ Nm} \quad \checkmark$$

Magnetic Torque:

$$T_{\text{mag}} = DB$$

$$\dots = D \frac{2M}{R^3} \text{ for pole (worst case)}$$

$$\dots = 0.2 \text{ A} \cdot \text{m}^2 \cdot \frac{2 \cdot 7.96 \cdot 10^{15} \text{ T} \cdot \text{m}^3}{(6578000)^3 \text{ m}^3}$$

$$\dots = 1.1186 \cdot 10^{-5} \text{ Nm} \quad \checkmark$$

Solar Radiation Pressure:

$$F_{\text{SRP}} = \frac{S}{c} A_s (1 + q) \cos(i)$$

$$\dots = \frac{1367}{3 \cdot 10^8} (0.04) (1 + 0.6) \cos(0^\circ)$$

$$\dots = 2.9163 \cdot 10^{-7}$$

$$T_{\text{SRP}} = F_{\text{SRP}} (c_{ps} - c_g)$$

$$\dots = 2.9163 \cdot 10^{-7} \cdot (0.06 \text{ m})$$

$$\dots = 1.74976 \cdot 10^{-8} \text{ Nm}$$

Drag:

$$F_{\text{drag}} = 0.5 \rho C_d A V^2 \quad \checkmark$$

We calculate ρ to be:

$$\rho = \exp(0.00001h^2 - 0.0274h - 16.947) \frac{\text{kg}}{\text{m}^3}$$

$$\rho_{200 \text{ km}} = \exp(0.00001(200)^2 - 0.0274(200) - 16.947) \frac{\text{kg}}{\text{m}^3}$$

$$\dots = 2.7152 \cdot 10^{-10} \quad \checkmark$$

We calculate V^2 for circular:

$$V^2 = \frac{\mu}{R}$$

$$\dots = \frac{3.986005 \cdot 10^{14}}{6578000}$$

$$\dots = 6.0596 \cdot 10^7 \quad \checkmark$$

Plugging back into F_{drag} :

$$F_{\text{drag}} = 0.5 \cdot 2.7152 \cdot 10^{-10} \cdot 1.2 \cdot 0.04 \cdot 6.0596 \cdot 10^7$$

$$\dots = 3.9487 \cdot 10^{-4}$$

Then into T_{drag} :

$$T_{\text{drag}} = F_{\text{drag}}(c_{pa} - c_g)$$

$$\dots = 3.9487 \cdot 10^{-4}(0.06)$$

$$= 2.3692 \cdot 10^{-5} \text{ Nm} \quad \checkmark$$

Adding:

$$T_{\text{total}} = T_{\text{grav}} + T_{\text{mag}} + T_{\text{SRP}} + T_{\text{drag}}$$

$$\dots = 8.4025 \cdot 10^{-9} + 1.1186 \cdot 10^{-5} + 1.74976 \cdot 10^{-8} + 2.3692 \cdot 10^{-5}$$

$$\dots = 3.4904 \cdot 10^{-5} \text{ Nm} \quad \checkmark$$

Given a margin of "1", we need to generate $3.4904 \cdot 10^{-5}$ Nm of torque

3.2 Angular Momentum Storage Requirement

Calculating the period for a 200 km altitude:

$$P = 2\pi \sqrt{\frac{a^3}{\mu}}$$

$$\dots = 2\pi \sqrt{\frac{6578^3}{398600.5}}$$

$$\dots = 5309.4771 \text{ seconds} \quad \checkmark$$

Multiply time by torque to get capacity required.

$$H_{\text{req}} \geq T_{\text{total}} \Delta t$$

$$\dots = 3.4904 \cdot 10^{-5} \cdot 5309.4771$$

$$\dots = 0.1853 \text{ Nms} \quad \checkmark$$

3.3 Torque for Slew

For a 180 degree turn in three seconds:

$$T_{\text{RW,Slew}} = 4\theta \frac{I}{t_{\text{slew}}^2}$$

$$\dots = 4\pi \frac{0.014}{3^2}$$

$$\dots = 0.01955 \text{ Nm} \quad \checkmark$$

4 Torque Wheel Capacity

4.1 Max Torque Production

Torque is just moment of inertia times angular acceleration

$$T_{\max} = I\alpha_{\max}$$

$$\dots = 0.0035 \cdot (50 \cdot 2\pi/60)$$

$$\dots = 0.018326 \text{ Nm} \quad \checkmark$$

Beware RPMs, for they are not radians per second.

4.2 Angular Momentum Storage

Angular momentum is moment of inertia times maximum angular velocity

$$H_{\max} = I\omega_{\max}$$

$$\dots = 0.0035 \cdot (410 \cdot 2\pi/60)$$

$$\dots = 0.1503 \text{ Nms} \quad \checkmark$$

5 Design Margins

5.1 Margins

- Disturbance Torque: **Satisfactory**
 - We can provide 0.018326 Nm of torque
 - We require $3.4904 \cdot 10^{-5}$ Nm
 - We have 0.01829 Nm to **spare**
- Angular Momentum Storage: **Unsatisfactory**
 - We can provide 0.1503 Nms of momentum
 - We require 0.1853 Nms
 - We are 0.0350 Nms of **deficit**
- Slew Rate: **Unsatisfactory**
 - We can provide 0.018326 Nm of torque
 - We require 0.01955 Nm
 - We are 0.001224 Nm of **deficit**

Proposal:

- Increase the max wheel speed by 30% to 533 rpm.

$$- H_{\max} = I\omega_{\max} = 0.0035 \cdot (574 \cdot 2\pi/60) = 0.1953 \text{ Nms}$$

- * This is now a 0.0100 Nms margin. Satisfies angular momentum storage

- Adjust shape/mass distribution of KestrelSat to reduce moment of inertia. Decrease moment of inertia to 0.013 kg m^2 .

$$- T_{\text{RW,Slew}} = 4\theta \frac{I}{t_{\text{slew}}^2} = 4\pi \frac{0.013}{3^2} = 0.01815 \text{ Nm}$$

- * This is now a 0.0001746 Nm margin. Satisfies slew rate

Rationale:

For angular momentum storage, I am hesitant to increase moment of inertia of wheel because that will either increase the size of the wheel or make it heavier, neither of which are great options

For slew rate, I would rather make the satellite lighter than make the reaction wheel bigger (and more expensive)

GREAT

